

4 (a) (i) *stress* is force / area B1 [1]

(ii) *strain* is extension / original length B1 [1]

(b) (i) $E = [F / A] \div [e / l]$ C1
 $e = (25 \times 1.7) / (5.74 \times 10^{-8} \times 1.6 \times 10^{11})$ C1
 $e = 4.6 \times 10^{-3} \text{ m}$ A1 [3]

(ii) A becomes $A/2$ or stress is doubled B1
 $e \propto l / A$ or substitution into full formula B1
total extension increase is $4e$ A1 [3]

5 (a) (i) $I = 12 / (6 + 12)$ C1